Conditional Probability and Bayes' Rule

50 minutes • 50 points • Instructor key

Grading note. Subpart scores sum to the problem totals. Award method credit for correct reasoning after an earlier error; do not deduct twice for the same arithmetic mistake. Suggested working time is 46 minutes, leaving 4 minutes to check answers.

Instructions. Show your reasoning and give exact fractions. No calculator is needed. Show the Bayes' Rule setup in Problems 3(c) and 4(d). The suggested times leave 4 minutes to check your work.

1. Warm-up: a finite probability space

8 points · about 7 minutes

A bag contains eight tickets labeled a, b, b, c, c, d, d, d. Choose one ticket uniformly at random and record its letter. An event is any subset of the sample space of recorded letters.

(a) Give the sample space Ω of recorded letters and the probability of each outcome. Check that these probabilities sum to 1. (4 points)

Solution. $\Omega = \{a, b, c, d\}$, with $\mathbb{P}(\{a\}) = 1/8$, $\mathbb{P}(\{b\}) = 2/8$, $\mathbb{P}(\{c\}) = 2/8$, and $\mathbb{P}(\{d\}) = 3/8$. The masses are nonnegative and sum to $(1 + 2 + 2 + 3)/8 = 1$. For any event, add the masses of its letters; disjoint events have no letters in common, so their probabilities add.

Credit: Sample space: 1; outcome probabilities: 2; sum check: 1.

(b) Let $A = \{a, c\}$ and $B = \{c, d\}$. Find $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cup B)$. (2 points)

Solution. $A \cap B = \{c\}$, so $\mathbb{P}(A \cap B) = 2/8 = 1/4$. Also $A \cup B = \{a, c, d\}$, so $\mathbb{P}(A \cup B) = (1 + 2 + 3)/8 = 3/4$.

Credit: Each event probability: 1.

(c) A student claims $\mathbb{P}(A) = 2/4$ because A contains two of the four possible letters. Is this correct? Give $\mathbb{P}(A)$ and explain. (2 points)

Solution. No: $\mathbb{P}(A) = (1 + 2)/8 = 3/8$. The tickets are equally likely, but the recorded letters are not. Counting letters and dividing by four would require all four letters to be equally likely.

Credit: Correct probability: 1; unequal letter probabilities: 1.

2. Which group is the denominator?

10 points · about 10 minutes

The table describes a group of 60 students. Choose one of them uniformly at random. Let C mean "belongs to the chess club" and M mean "plays a musical instrument."

	M	M^c	Total
C	12	18	30
C^c	8	22	30
Total	20	40	60

(a) Find $\mathbb{P}(C)$, $\mathbb{P}(M)$, and $\mathbb{P}(C \cap M)$. (3 points)

Solution. $\mathbb{P}(C) = 30/60 = 1/2$, $\mathbb{P}(M) = 20/60 = 1/3$, and $\mathbb{P}(C \cap M) = 12/60 = 1/5$.

Credit: Each probability: 1.

(b) Find $\mathbb{P}(M \mid C)$ and $\mathbb{P}(C \mid M)$. For each calculation, say which group supplies the denominator. (4 points)

Solution. $\mathbb{P}(M \mid C) = 12/30 = 2/5$: the denominator counts the 30 chess-club members. $\mathbb{P}(C \mid M) = 12/20 = 3/5$: the denominator counts the 20 students who play an instrument.

Credit: Each probability: 1; each conditioning group: 1.

(c) Are C and M independent? Justify your answer using probabilities. (3 points)

Solution. No: $\mathbb{P}(C \cap M) = 1/5$, whereas $\mathbb{P}(C)\mathbb{P}(M) = (1/2)(1/3) = 1/6$. Equivalently, $\mathbb{P}(M \mid C) = 2/5 \neq 1/3 = \mathbb{P}(M)$.

Credit: A valid independence criterion: 1; numerical comparison: 1; conclusion: 1. Reversing the condition and getting different values is not by itself an independence test.

3. A red ball: which box?

12 points · about 11 minutes

Box U contains three red balls and one blue ball. Box V contains one red ball and three blue balls. Choose box U with probability $1/3$ and box V with probability $2/3$. Then draw one ball uniformly at random from the chosen box. Let U and V also denote the events that the corresponding box is chosen, and let R denote the event that the ball is red.

(a) List the four possible outcomes of the form (chosen box, ball color) and find their probabilities. Show how the probabilities for box choice and color combine. A labeled tree or table is fine. (4 points)

Solution. The conditional red/blue probabilities are $3/4, 1/4$ in box U and $1/4, 3/4$ in box V. Multiply along each path:

Chosen box	Color	Joint probability
U	Red	$(1/3)(3/4) = 1/4$
U	Blue	$(1/3)(1/4) = 1/12$
V	Red	$(2/3)(1/4) = 1/6$
V	Blue	$(2/3)(3/4) = 1/2$

These four disjoint outcomes have total probability $1/4 + 1/12 + 1/6 + 1/2 = 1$.

Credit: Each outcome with its product and probability: 1.

(b) Use the law of total probability to find $\mathbb{P}(R)$. (3 points)

Solution. The box choices U, V are disjoint and cover all possibilities, so

$$\mathbb{P}(R) = \mathbb{P}(R \mid U)\mathbb{P}(U) + \mathbb{P}(R \mid V)\mathbb{P}(V) = \frac{3}{4} \cdot \frac{1}{3} + \frac{1}{4} \cdot \frac{2}{3} = \frac{5}{12}.$$

Credit: Decomposition into the two box cases: 1; correct weighted terms: 1; result: 1.

(c) The drawn ball is red. Use Bayes' Rule to find the probability that box U was chosen. (3 points)

Solution.

$$\mathbb{P}(U \mid R) = \frac{\mathbb{P}(R \mid U)\mathbb{P}(U)}{\mathbb{P}(R)} = \frac{(3/4)(1/3)}{5/12} = \frac{1/4}{5/12} = \frac{3}{5}.$$

Credit: Bayes setup with the prior in the numerator: 1; correct substitution (or consistent use of part (b)): 1; evaluation: 1.

(d) Another student answers part (c) with $3/4$, saying "three of the four balls in box U are red." Which conditional probability does $3/4$ describe, and how does it differ from the probability requested? (2 points)

Solution. $3/4$ is $\mathbb{P}(R \mid U)$: the probability of red when the box is known to be U. The question asks for $\mathbb{P}(U \mid R)$: the probability the box was U when the color is known to be red. The box choice is uncertain, and both boxes can produce a red ball.

Credit: Identify $\mathbb{P}(R \mid U)$: 1; explain the reversed conditioning: 1.

4. Monty Hall, step by step

20 points · about 18 minutes

A car is placed behind one of three doors, with probability 1/3 for each door. The other two doors hide goats. You initially choose Door 1. The host follows these rules:

- The host knows where the car is and always opens exactly one unchosen door that hides a goat.
- If both unchosen doors hide goats, the host chooses between them with equal probability 1/2.
- The host always offers you the choice of staying with Door 1 or switching to the other unopened door.

You observe the host open Door 3, revealing a goat. Let C_i mean "the car is behind Door i," and let H_3 mean "the host opens Door 3." All probabilities below assume your initial choice is Door 1.

(a) Before the host opens a door, what are $\mathbb{P}(C_1)$, $\mathbb{P}(C_2)$, and $\mathbb{P}(C_3)$? (2 points)

Solution. $\mathbb{P}(C_1) = \mathbb{P}(C_2) = \mathbb{P}(C_3) = 1/3$.

Credit: Equal probabilities: 1; correct value 1/3: 1.

(b) Fill in the table and briefly explain each entry using the host's rules. These are probabilities of the host's action *given the car's location.* (6 points)

Car location	C_1	C_2	C_3
$\mathbb{P}(H_3 \mid C_i)$	1/2	1	0

Solution. If C_1 occurs, Doors 2 and 3 both hide goats, so the host chooses Door 3 with probability 1/2. If C_2 occurs, the host must open Door 3. If C_3 occurs, opening Door 3 would reveal the car and is forbidden by the rules.

Credit: Each conditional probability: 1; each explanation: 1.

(c) Use the law of total probability to find $\mathbb{P}(H_3)$. Show all three car-location terms. (4 points)

Solution. The three car locations form disjoint cases covering all possibilities:

$$\mathbb{P}(H_3) = \sum_{i=1}^{3} \mathbb{P}(H_3 \mid C_i)\mathbb{P}(C_i) = \frac{1}{2}\cdot\frac{1}{3} + 1\cdot\frac{1}{3} + 0\cdot\frac{1}{3} = \frac{1}{2}.$$

Credit: Total-probability setup: 1; three weighted terms: 2; evaluation: 1.

(d) Use Bayes' Rule to find $\mathbb{P}(C_1 \mid H_3)$ and $\mathbb{P}(C_2 \mid H_3)$. Also state $\mathbb{P}(C_3 \mid H_3)$. (5 points)

Solution.

$$\mathbb{P}(C_i \mid H_3) = \frac{\mathbb{P}(H_3 \mid C_i)\mathbb{P}(C_i)}{\mathbb{P}(H_3)}.$$

Thus

$$\mathbb{P}(C_1 \mid H_3) = \frac{(1/2)(1/3)}{1/2} = \frac{1}{3}, \qquad \mathbb{P}(C_2 \mid H_3) = \frac{1(1/3)}{1/2} = \frac{2}{3}.$$

$\mathbb{P}(C_3 \mid H_3) = 0$, since Door 3 has revealed a goat. The three updated probabilities sum to 1.

Credit: For each of Doors 1 and 2, Bayes setup: 1 and calculation: 1; Door 3 probability: 1.

(e) Should you stay with Door 1 or switch to Door 2? Give the chance of winning under each choice. Explain why "two doors remain, so each has probability 1/2" is not a valid argument. (3 points)

Solution. Switch to Door 2: switching wins with probability 2/3, while staying wins with probability 1/3, given H_3. Two remaining possibilities need not be equally likely. Opening Door 3 is twice as likely when the car is behind Door 2 as when it is behind Door 1; with equal priors, Bayes' Rule therefore gives Door 2 twice the updated probability.

Credit: Switch decision: 1; both winning probabilities: 1; explanation referring to unequal likelihoods or the host's informed rule: 1.